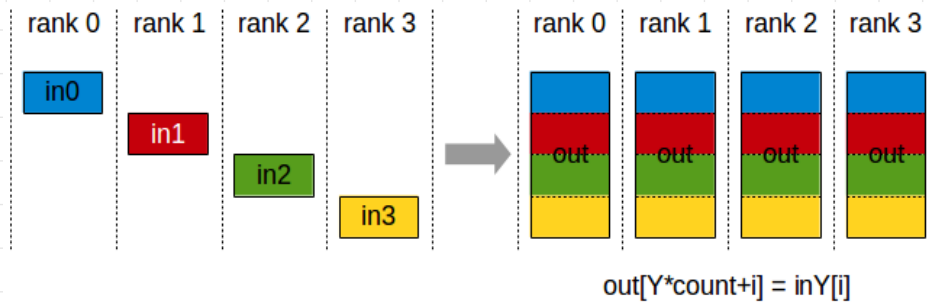
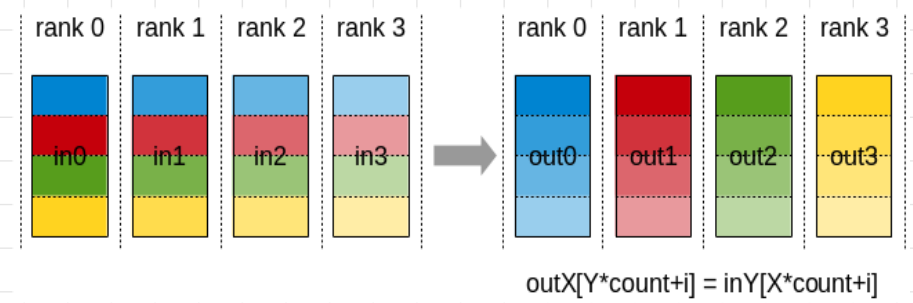


Collectives

All-Gather (AG)

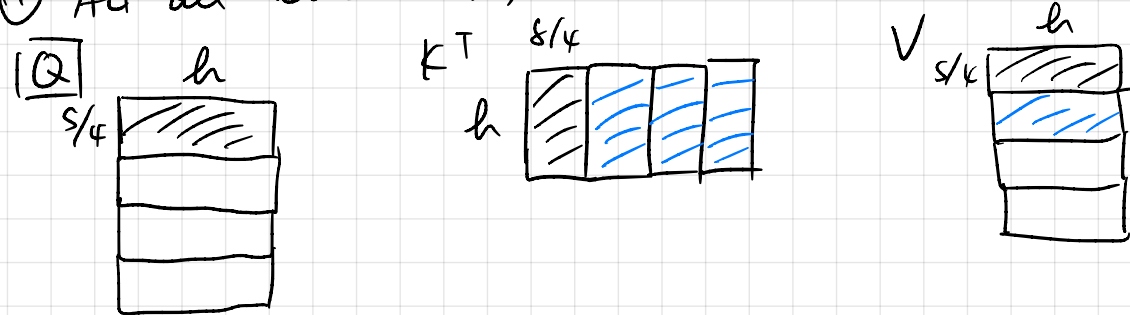


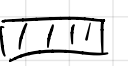
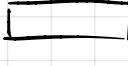
All-to-all (A2A)

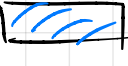


Meta's all-gather-based context parallelism

① AG all the KV, now each rank has



( : supposed to be on rank 0;  : not on rank 0 after AG

 : transferred to rank 0 after AG) Attn map's 1/4

$$QK^T \Rightarrow (\frac{s}{4}, h) \times (h, s) \Rightarrow (\frac{s}{4}, s) \rightarrow \text{Attn map's } 1/4$$

$$\text{sm}\left(\frac{QK^T}{\sqrt{d}}\right)V \Rightarrow (\frac{s}{4}, s) \times (s, h) \Rightarrow (\frac{s}{4}, h) \rightarrow \text{Final result's } 1/4$$

Context Parallelism

Self-attention : $\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$

$Q \in [s, h]$
 $\downarrow$ seq. len. $\downarrow$ hidden size

$K \in [s, h]$

$V \in [s, h]$

Assume 4 GPUs (rank 0~3)

Each GPU has

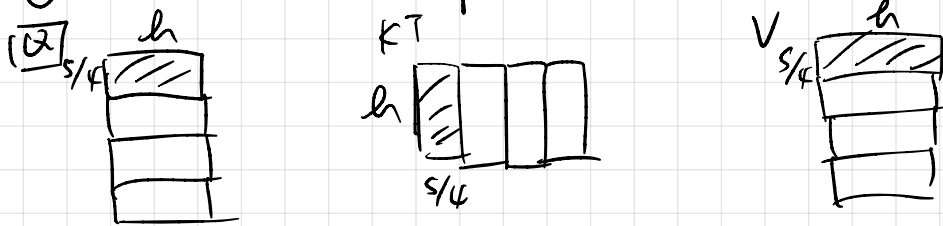
$Q \in [\frac{s}{4}, h]$

$K \in [\frac{s}{4}, h]$

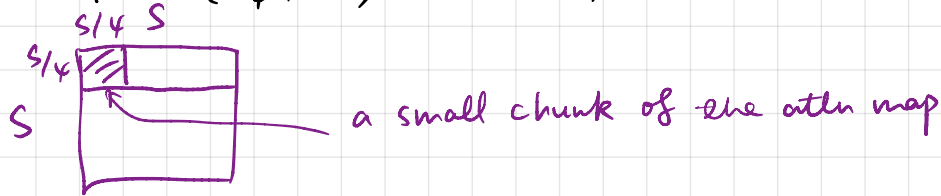
$V \in [\frac{s}{4}, h]$

A2A-based CP (a.k.a. ring attention)

① Each rank computes its own chunk

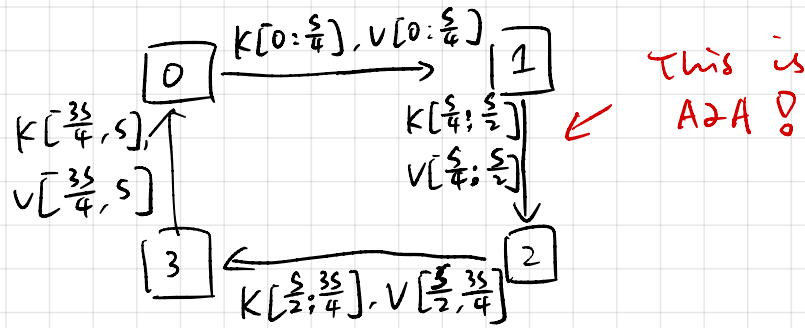


$$QK^T: \left(\frac{s}{4}, h\right) \times \left(h, \frac{s}{4}\right) \Rightarrow \left(\frac{s}{4}, \frac{s}{4}\right)$$

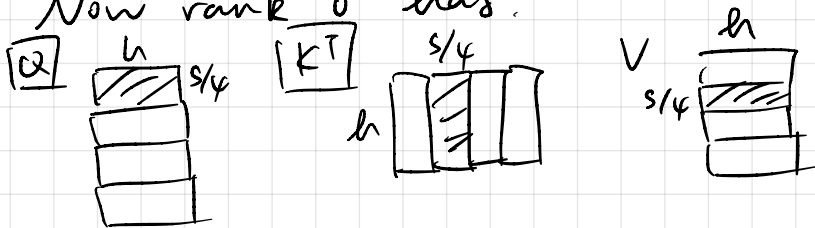


② For all rank i :

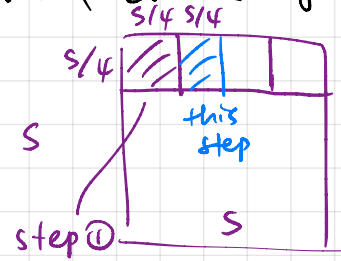
send the current K, V chunk to $(i+1) \% 4$ // suppose 4 ranks



Now rank 0 has:



③ The second step's computation produces another small chunk of the attn map



(and then repeat for 4 steps)

Questions:

① Why is A2A-based CP not popular today?

② How to calculate softmax for ring attention?

Note that softmax needs the whole row in attn map!

